

RISK SCORING & PRIORITIZATION SYSTEM

“An Explainable Machine Learning Framework for
Anti-Money Laundering Risk Assessment and
Investigation Prioritization”

Technical Documentation

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1. Executive Summary

This project addresses the challenge of risk scoring and prioritization in anti-money laundering (AML) investigations. The objective is not merely to classify transactions as suspicious or non-suspicious, but to generate an actionable account-level risk score that helps analysts prioritize investigations efficiently.

The proposed solution combines machine learning, behavioral analysis, account-level aggregation, and explainable AI to identify high-risk accounts and provide transparent reasoning behind each assessment.

2. Problem Understanding

Financial institutions process a large volume of transactions daily, making it impractical for analysts to investigate every account equally. Traditional rule-based systems often generate excessive alerts, increasing operational burden and reducing investigation efficiency.

Traditional rule-based systems typically use static thresholds (e.g., flag all transactions > \$10,000). This approach fails because:

- Thresholds are easily circumvented by sophisticated actors
- Context is ignored—a \$9,000 transaction may be riskier than a \$50,000 one if it involves a high-risk jurisdiction
- Alert volume is unmanageable—investigators receive thousands of alerts daily, most of which are false positives

To address this challenge, we developed a risk scoring framework that:

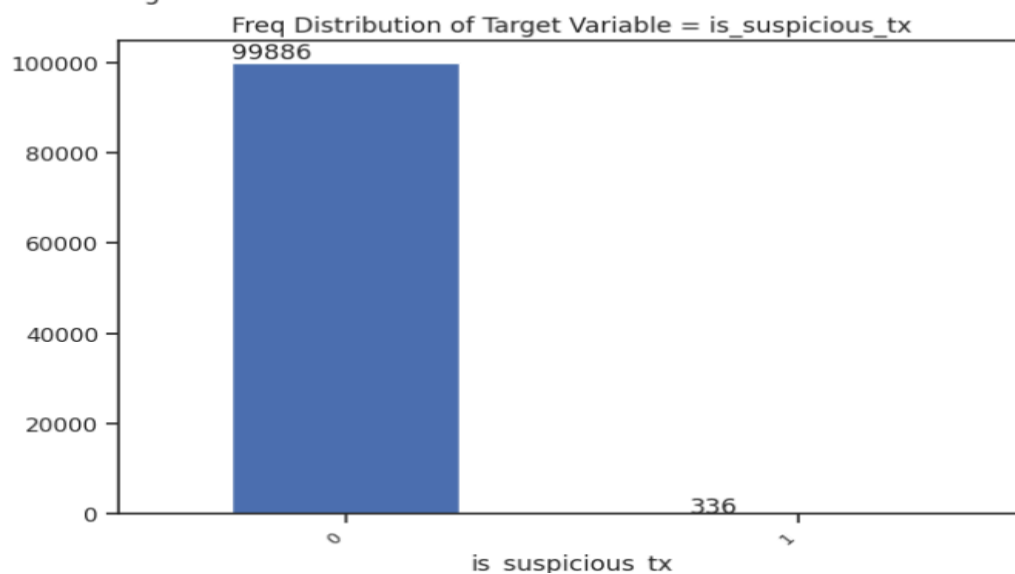
- Estimates the probability of suspicious activity
- Aggregates transaction-level signals into account-level intelligence
- Produces a ranked investigation queue
- Explains the drivers behind each risk score

3. Data Understanding and Exploratory Analysis

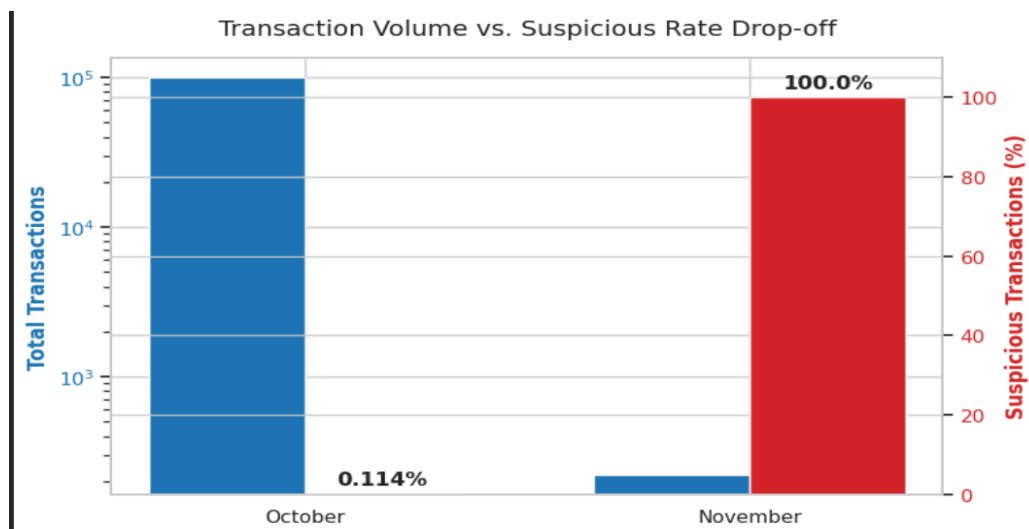
The dataset consists of transaction-level records containing transactional, behavioral, and KYC-related features.

Key observations from exploratory analysis included:

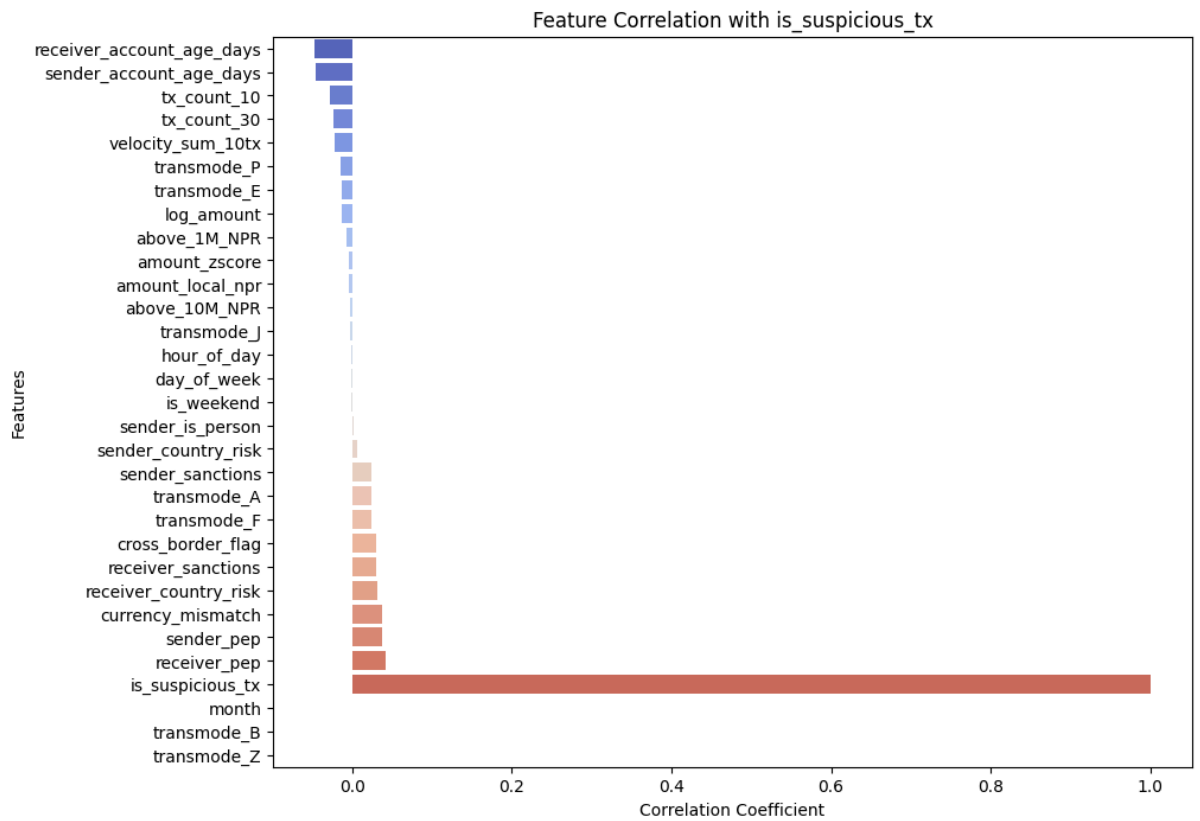
- Significant class imbalance, with suspicious transactions representing only a small fraction of total records. (Nearly 99.3% of the data were not suspicious and 0.7% of data were suspicious)



- There was a spike rise in the fraud transaction were performed in the month of the November



- Strong behavioral indicators such as transaction velocity and amount variance.
- Elevated risk patterns among accounts associated with PEP and sanctions indicators.
- A correlation was used to determine how numeric values were correlated with the target.



4. Feature Engineering

Feature engineering focused on capturing meaningful AML risk indicators rather than relying solely on raw transaction attributes.

Following feature Engineering were develop for enhancing the model

- Amount-to-Average Ratio —calculate the ratio of the current amount transaction to average amount transaction.
- Amount-to-Max Ratio — calculate the ratio of the current amount transaction to max amount transaction.
- Country Risk Interaction — multiply sender and receiver country risk to highlight dangerous corridors.
- Cross-Border High-Risk Flag — cross-border AND at least one high-risk country involved.
- PEP/Sanctions Threat — anyone in the transaction (either side) flagged as a PEP or under sanctions.

These features were designed to improve model sensitivity to unusual transaction behavior.

Account-Level Aggregation

We aggregated transaction-level signals to account-level profiles using a mix of statistical functions:

Aggregation Type	Examples	Purpose
Count	Total transactions	Captures activity volume
Sum	Total amount transacted	Identifies high-value accounts
Mean	Average transaction amount, average risk scores	Captures typical behavior
Max	Maximum transaction amount, maximum receiver risk	Identifies worst-case exposures
Std	Standard deviation of amounts, hours, velocities	Captures behavioral variability
Nunique	Unique receiver accounts	Detects smurfing patterns

5. Model Development

Given the highly imbalanced nature of AML data, tree-based ensemble models were selected due to their robustness and ability to capture non-linear relationships.

Models evaluated:

- Random Forest
- LightGBM
- Artificial Neural Network(ANN)

Model performance was assessed using stratified validation and metrics appropriate for imbalanced classification.

LightGBM was selected for the following technical reasons:

- Native Support for Imbalanced Classification: The `scale_pos_weight` parameter enables direct adjustment for class imbalance, avoiding manual oversampling or SMOTE.
- Fast Inference: Critical for production deployments where thousands of accounts need scoring daily.
- Robustness to Overfitting: With appropriate regularization (`reg_alpha`, `reg_lambda`, `max_depth`), LightGBM generalizes well to unseen data.

6. Hyperparameter Tuning

We optimized hyperparameters specifically for ranking quality rather than classification accuracy. To maximize the predictive performance of our model, we employed Randomized Hyperparameter Tuning (RandomizedSearchCV). Models contain critical parameters—such as tree depth, learning rate, and regularization coefficients—that must be set before training begins. Finding the optimal combination of these settings is essential for maximizing metric performance (like PR-AUC and Precision@K) and preventing overfitting.

Why Randomized Search Was Selected Over Grid Search?

While traditional Grid Search performs an exhaustive sweep over all possible parameter combinations, it scales poorly and is computationally expensive. Randomized Hyperparameter Tuning solves this by sampling a fixed number of parameter combinations from specified probability distributions. This approach is mathematically proven to find optimal or near-optimal configurations in a fraction of the time, allowing us to evaluate a much broader search space within our compute budget.

After tuning all the parameter the best outcome we get are:

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{  
  "objective": "binary",  
  "metric": "average_precision",  
  "boosting_type": "gbdt",  
  "learning_rate": 0.015,  
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  "colsample_bytree": 0.65,  
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  "verbose": -1,  
  "n_estimators": 800  
}
```

7. Account-Level Risk Scoring

A key contribution of this project is the transition from transaction-level predictions to account-level intelligence.

The trained model produces a risk probability for each transaction. These probabilities are then aggregated to construct an account-level risk profile.

The resulting risk score reflects:

- Historical transaction behavior
- Transaction frequency and velocity
- Risk concentration across multiple transactions
- KYC-related risk indicators

This enables direct comparison and prioritization of accounts.

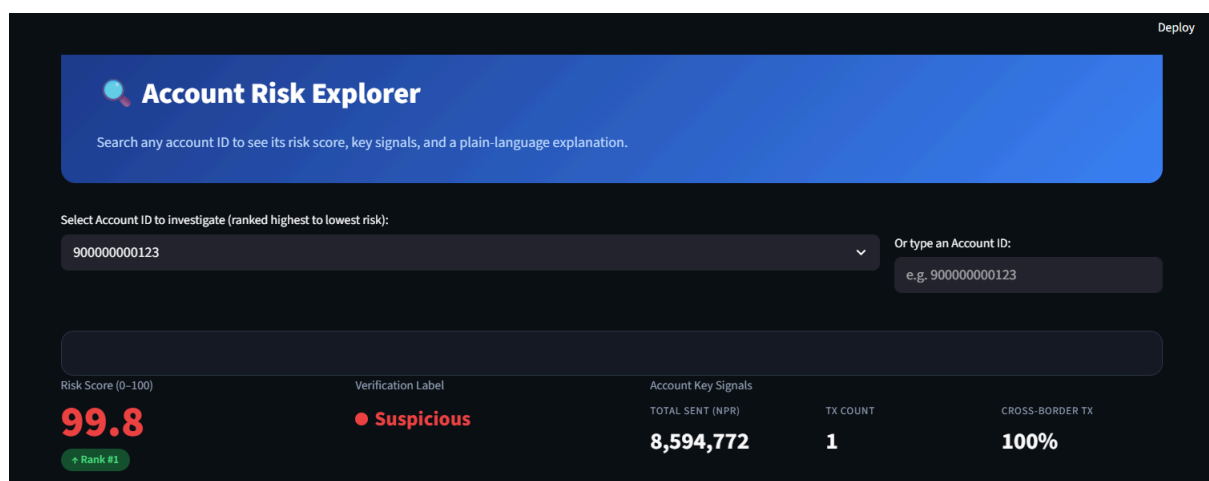
8. Risk Prioritization Framework

Accounts are ranked according to their calculated risk scores. The output is a prioritized investigation queue that allows analysts to focus resources on the highest-risk accounts first. Unlike traditional classification systems, this ranking-oriented approach directly supports operational decision-making.

The problem statement explicitly requires risk scores and ranked lists, not binary predictions. We designed our approach around this requirement:

Aspect	Classification Approach	Our Ranking Approach
Output	Binary (0/1)	Continuous probability (0-1)
Evaluation	Accuracy, F1, AUC	Precision@K, Recall@K
Business Value	Alert generation	Investigation queue prioritization
Investigator Workflow	Review all alerts	Review top K accounts, ignore low-risk

Key technical insight: Even if classification accuracy is high, the ranking quality may be poor. For example, a model with 99% accuracy could still place suspicious accounts at positions 1000+, making them effectively invisible to investigators. Our focus on Precision@K ensures that the highest-risk accounts actually appear at the top of the ranked list.



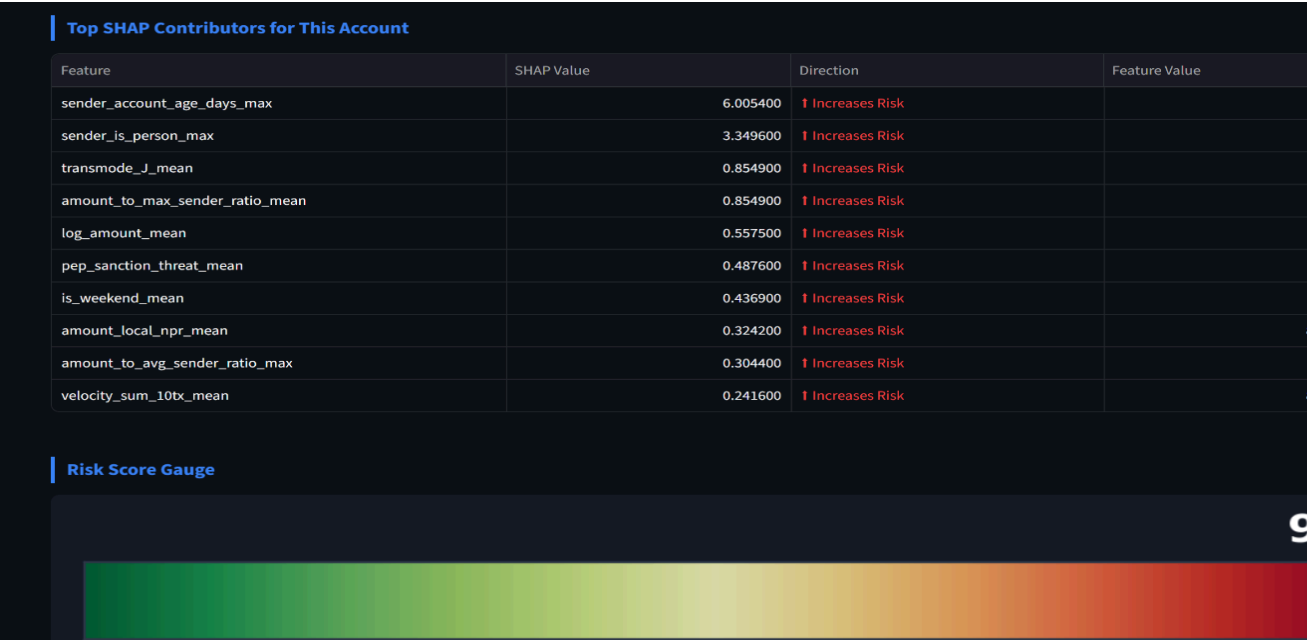
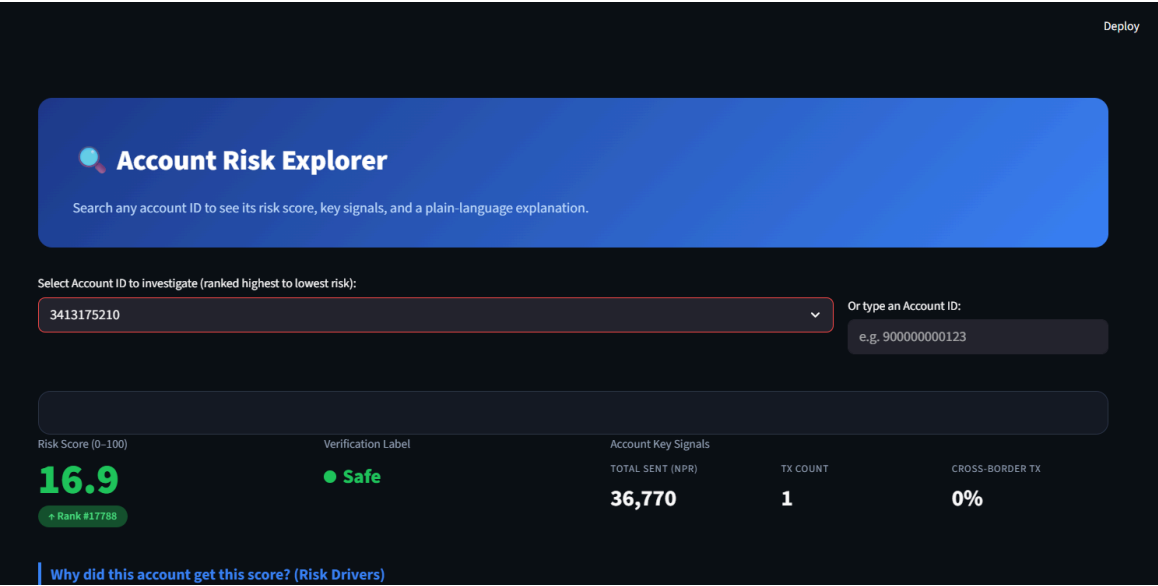


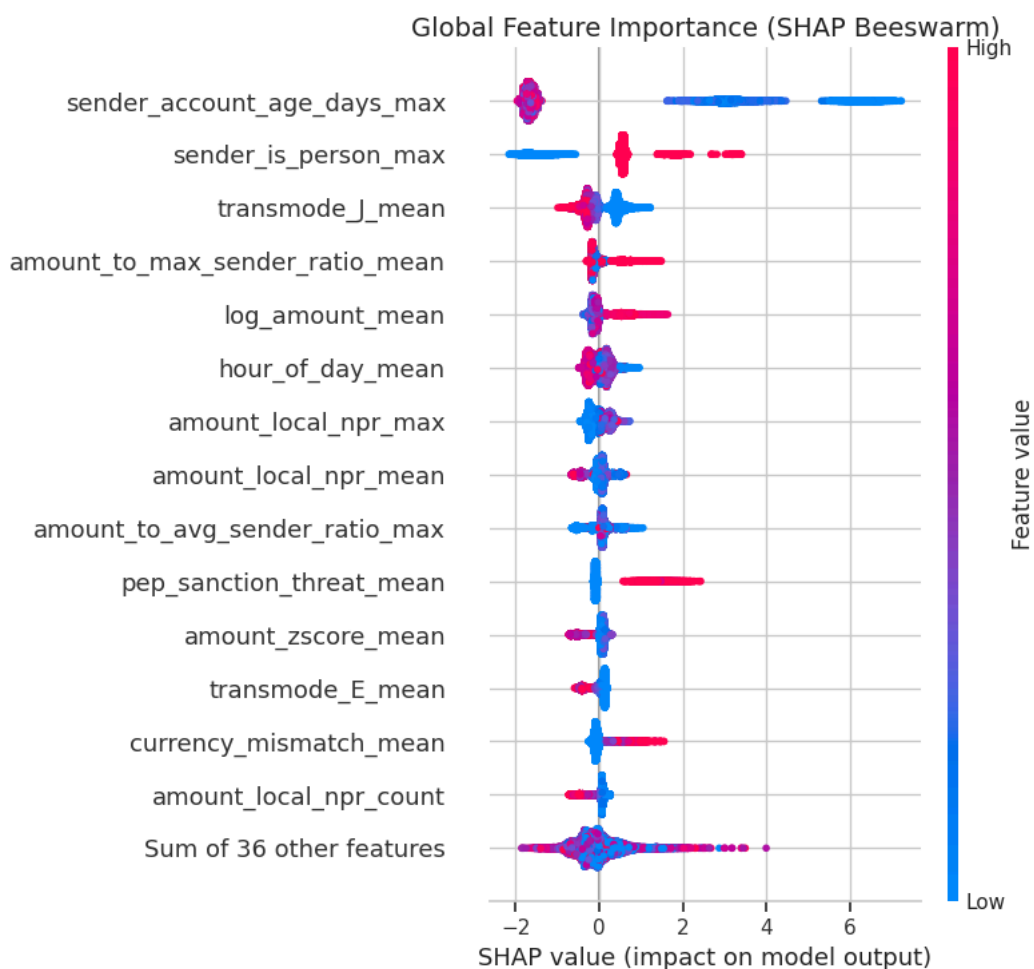
Fig Shap analysis for each account

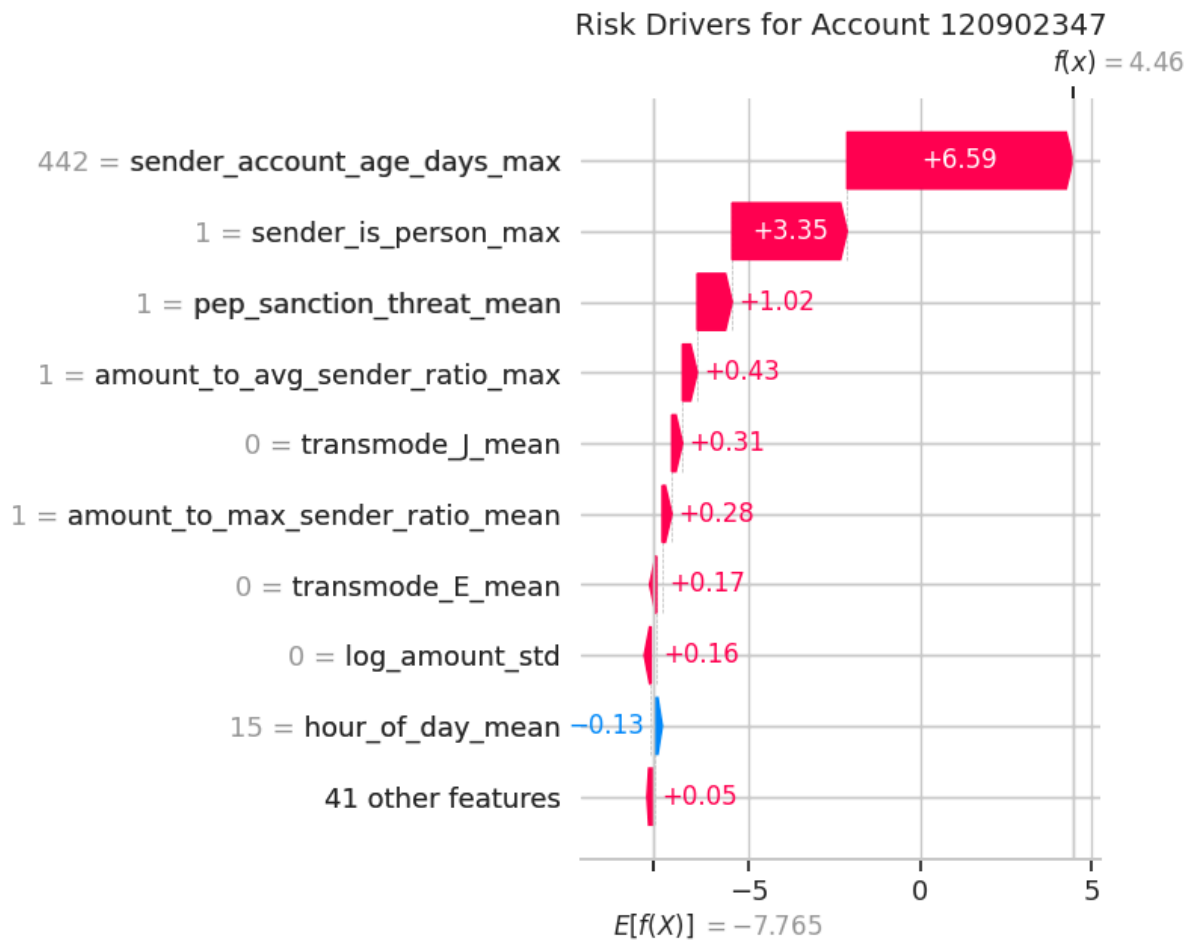
9. Explainable AI

Trust and transparency are critical in financial crime investigations. SHAP (SHapley Additive Explanations) was incorporated to explain model predictions and identify the factors contributing most strongly to each risk assessment. For every high-risk account, analysts can view the primary drivers behind the assigned score, improving auditability and confidence in the system.

Top 3 Global Drivers:

- sender_account_age_days (Account age)—New accounts are disproportionately high-risk
- pep_sanction_threat_mean (PEP/Sanctions involvement)—Critical red flag
- amount_to_max_sender_ratio_mean (Amount deviation)—Unusual transaction sizes





Our model flags suspicious financial activity by analyzing user behavior. The most critical red flag is a brand-new account profile, which immediately shifts the model toward a high-risk prediction. Conversely, established, mature accounts are consistently treated as safe and normal. While the model also evaluates over 40 minor details—like specific transaction types, mismatched currencies, and sanction threats—most of these have zero impact on the average user. Instead, they act as a safety net, triggering an alert only when a rare, highly specific combination of minor anomalies occurs simultaneously.

10. Evaluation Strategy

Suspicious accounts are a tiny fraction of the total data (only about 1%). Because of this, looking at "Overall Accuracy" does not work. A model could simply guess that everyone is innocent and be 99% accurate, while missing every single criminal. Therefore, we use specific metrics that focus on finding the real bad actors without creating a mountain of false alarms.

We measure success using four main criteria:

Metric	What It Means	Why It Matters
ROC-AUC	Overall grading power	Shows how well the model separates good accounts from bad ones across the entire system.
PR-AUC	Accuracy on rare events	Proves if the model is actually good at finding rare fraud, rather than just guessing "innocent."
Recall	Catch rate	Measures what percentage of the total 237 suspicious accounts we successfully found.

Precision@K Accuracy in the top spots Measures exactly how many true criminals are hiding in the very top "K" (e.g., top 10, 20, or 50) alerts.

Why We Care Most About Precision@K ?

We focus heaviest on Precision@K because investigators have limited time. They cannot check thousands of clean accounts looking for a needle in a haystack. By making sure the very top of the list is packed with actual suspicious accounts, investigators can catch fraud immediately on their first few reviews. This eliminates wasted hours and stops alert fatigue before it starts.

11. Performance Result

Metric	Value
ROC-AUC	0.9927
PR-AUC	0.8464
Suspicious Accounts	237/22,310 (1.06%)

Precision@K Analysis

K	Precision@K	Recall@K	Suspicious Caught
10	100.00%	4.22%	10/237
20	100.00%	8.44%	20/237
50	100.00%	21.10%	50/237
100	97.00%	40.93%	97/237
200	89.50%	75.53%	179/237
500	42.00%	88.61%	210/237

Interpretation: An investigator reviewing the top 50 accounts would encounter 100% suspicious accounts—every single investigation would be worthwhile. Reviewing the top 200 accounts would capture 75% of all suspicious accounts while maintaining 89.5% precision.

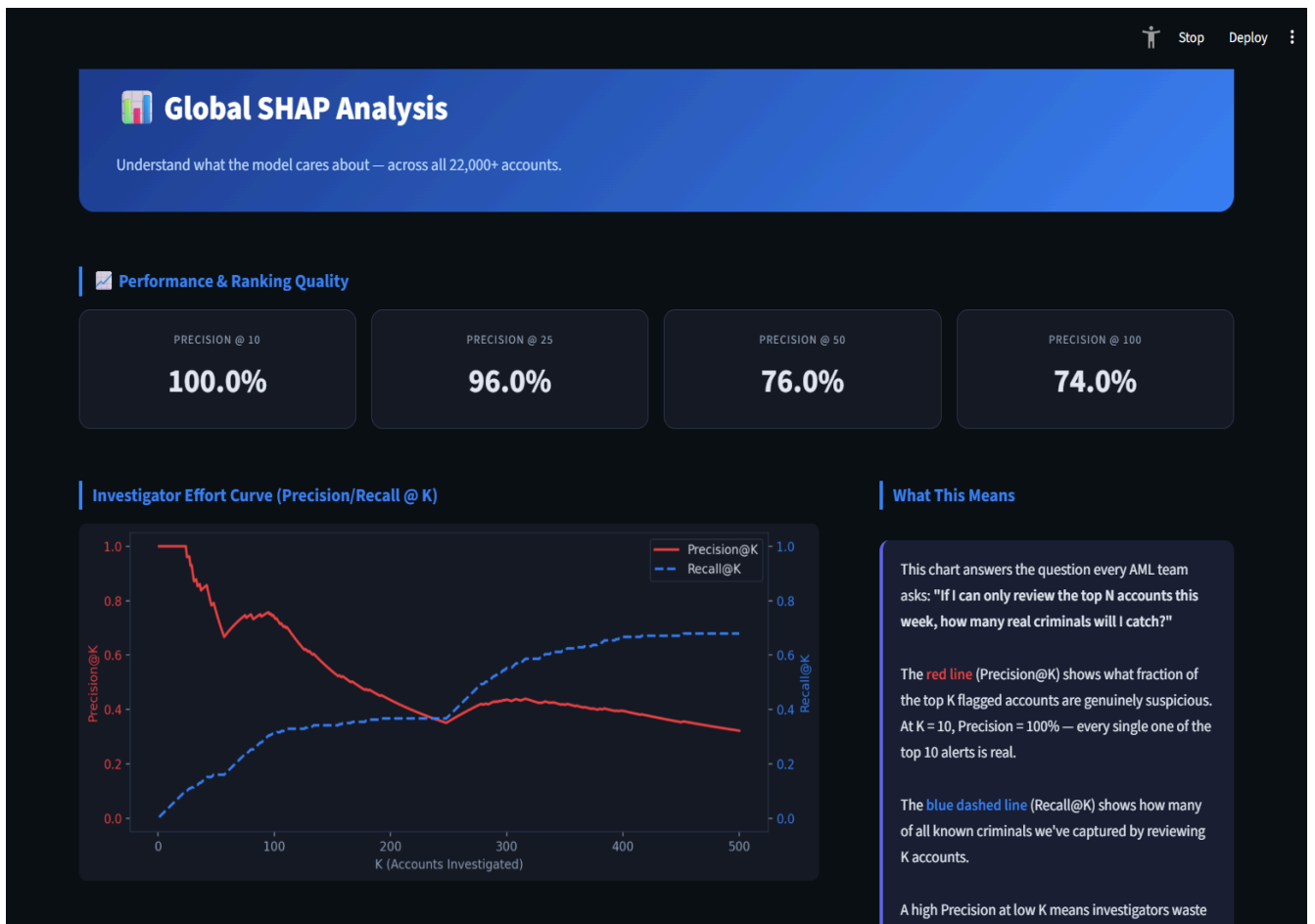


Fig: Dashboard Result

12. Conclusion

Our AML Risk Scoring & Prioritization Engine delivers:

- Accurate Risk Estimation: LightGBM ensemble model with ROC-AUC of 0.9927 and PR-AUC of 0.8464
- Actionable Ranking: Precision@50 = 100%, enabling investigators to focus on genuinely suspicious accounts
- Explainable Decisions: SHAP-based feature attribution with human-readable reasoning
- Production Readiness: Model persistence, batch inference, and metrics tracking
- Regulatory Compliance: Full auditability and transparency

13. References

- **LightGBM Documentation:** <https://lightgbm.readthedocs.io/>
- **SHAP Documentation:** <https://shap.readthedocs.io/>